

AapdaSetu

Offline Emergency Communication & Local AI Assistant

The Idea

AapdaSetu is an offline-first, peer-to-peer emergency communication system designed for situations where Internet connectivity is unavailable. Nearby devices discover each other over a local network and communicate directly without relying on a central server or cloud service.

The system will also provide an **offline AI emergency assistant**. A device running the local AI service can answer emergency-related questions using a locally stored knowledge base and a locally running language model. Other nearby devices can access that AI service through the P2P network.

What We Are Going to Implement

Component	Implementation
1. P2P Network	Devices communicate directly with each other without an Internet-based central server.
2. Automatic Peer Discovery	Nearby AapdaSetu nodes discover each other automatically on the same local network.
3. Emergency Chat	Users can send and receive messages between nearby devices.
4. Emergency Broadcast	A user can send an important/emergency message to connected peers.
5. Peer Status	The application shows nearby/connected peers and their connection status.
6. Offline AI Assistant	A local language model runs without external AI APIs or Internet access.
7. Emergency Knowledge Base	The AI retrieves information from locally stored emergency and safety material.
8. P2P AI Service	A device running the AI can expose it as a service that other nearby peers can query.
9. React Interface	A custom web interface provides chat, peer status, emergency broadcast, and AI interaction.
10. Containerized Deployment	The system will be packaged with Docker for reproducible local deployment.

Tech Stack

Layer	Technology
Frontend	React + TypeScript + Tailwind CSS
Backend / Core Node	Go
P2P Networking	libp2p
Peer Discovery	mDNS
Messaging	libp2p PubSub
Local API	Go HTTP API

AI Runtime	Ollama / llama.cpp
Local AI Model	Small quantized open-source LLM (e.g. Qwen or Llama family)
Knowledge Retrieval	Local RAG pipeline + local vector store
Containerization	Docker + Docker Compose
Testing	Go tests + frontend tests
CI/CD	GitHub Actions

Core Architecture

Nearby Devices → mDNS Discovery → libp2p Network → PubSub Messaging → Go Node → React UI

For AI-enabled nodes: **P2P Request → Local AI Service → Local RAG + Local LLM → P2P Response.**

Primary goal: build a genuinely offline-first emergency communication network where communication does not depend on the Internet, while providing optional local AI assistance.